DCIT 316 Final Project Proposal

Theme: Responsible Social Media Mining for Public Good under API Constraints

1. Project Title

Can we trust the confidence? Per-class calibration of Twi and Hausa hate-speech classifiers under class imbalance, using AfriHate

2. Problem Definition

A classifier attaches a confidence score to every prediction. Accuracy tells you how often it is right; it does not tell you whether that confidence can be believed. A model is calibrated when its confidence matches its real hit rate — of all posts it marks at 0.9 confidence, about 90% are correct. A model can be accurate yet miscalibrated: confident in a way its actual error rate does not earn. This gap is invisible in any single prediction and only appears when predictions are pooled and checked in aggregate.

This matters the moment a model is placed in front of a human. A moderation-triage tool ranks posts by predicted toxicity and asks a moderator to review the uncertain calls and trust the confident ones. That division of labour only works if confidence tracks correctness. If the model is confident on its errors, the tool files those errors under "trust me" and routes the moderator's attention away from exactly the mistakes they need to catch. It does not merely fail to help — it systematically hides errors in the pile the moderator was told to skip. It blinds the moderator.

Low-resource languages are where this is most likely to bite: models are weaker on them, and calibration is known to degrade under class imbalance (Islam et al., 2021) and for low-resource, typologically distant languages (Ahuja et al., 2022). Whether Twi hate-speech classifiers are calibrated well enough to support a triage tool is unknown. This project investigates whether a hate-speech model's confidence is equally trustworthy across classes and across two languages with inverted class balance.

- **Task:** supervised classification (Hate / Abusive / Normal — the dataset's label strings), analyzed for per-class calibration.
- **Unit of analysis:** the post.
- **Affected:** Twi and Hausa speakers subject to automated moderation — under-protected if hate is missed, over-censored if ordinary (Normal) speech is false-flagged; and teams deploying such tools without checking whether confidence can be trusted.
- **Useful evidence:** per-class Expected Calibration Error (ECE) and reliability diagrams for each model on each language, plus the fraction of the model's errors that fall in high-confidence bins (the "blinding" quantity). A finding that calibration is uniform across classes is a legitimate result and will be reported as one.

3. Motivation

Automated moderation of African-language content fails in both directions: hate speech goes undetected where models are weak, and ordinary speech is over-flagged where models are twitchy on text they poorly represent. AfriHate exists because these languages are both under-moderated and prone to keyword-driven over-censorship. Twi is among Ghana's most spoken languages and one of the smallest, most imbalanced subsets in that collection — 180 Normal posts against 3,278 Abusive. Hausa in the same collection is nearly inverted, with Normal the majority class. That contrast is the opportunity: the same model, the same task, two languages where the rare class swaps. If confidence is less trustworthy on whichever class is rare, the tool's failure mode flips between languages — missing hate in one, silencing speech in the other. Establishing whether that happens is directly useful to anyone building

moderation on African-language data, and it needs no data collection: two licensed, pre-annotated subsets and standard calibration tooling. It suits a final project because the contribution is in the measurement and the cross-language design, not in engineering.

4. Research Question

Is a hate-speech classifier's confidence equally trustworthy across classes, and does calibration on the minority class degrade — following the rare class when we compare Twi (rare: Normal) against Hausa (rare: Hate)?

Sub-questions: (a) Does a fine-tuned transformer, though more accurate, calibrate worse than a classical baseline? (b) Does post-hoc temperature scaling recover minority-class calibration, or only overall calibration?

5. Data Source and Access Method

Field	Value
Source	AfriHate — Twi (twi) and Hausa (hau) subsets
Access method	Approved public dataset (Hugging Face)
Link	huggingface.co/datasets/afrihate/afrihate
Reference	Muhammad et al. (2025), NAACL, 1854–1871, DOI 10.18653/v1/2025.naacl-long.92
Licence	**To be verified on the gated dataset page before use** — the HF card tags apache-2.0, the repository carries no license file, and the paper states none. Restricted to non-commercial academic use pending confirmation; recorded in the data card.
Unit	Post (tweet)
Sample size	Twi ≈ 3,901 (443 Hate / 3,278 Abusive / 180 Normal); Hausa ≈ 6,644 (343 Hate / 2,336 Abusive / 3,965 Normal) — to be confirmed on load
Raw data in GitHub?	**No** — derived outputs only (metrics, reliability diagrams, aggregate error categories)

No scraping, no API calls, no collection. Data is pre-annotated by native speakers and de-identified by its authors.

6. Ethics and Privacy Note

Personal data: Twitter-derived text may carry residual identifying content despite the authors' cleaning; posts also carry hate-target labels across sensitive attributes, encoding group information about real Ghanaian and Nigerian communities.

Mitigations: used as distributed — no re-identification, no user-level linkage, no tracing to accounts. No raw text committed; derived outputs only. No raw data uploaded to AI tools; AI assistance is limited to code scaffolding and writing, and is disclosed. Illustrative examples in the report are limited to a handful, redacted of handles, URLs, and names.

Harm if misinterpreted: the models here are weak and are diagnostic instruments, not deployable moderation systems. Two specific harms to name: reporting low Twi performance could be misread as Twi being intrinsically hard, and a triage tool trusting confident errors could either let hate through or silence ordinary speech. The project's point is precisely to surface the second risk before anyone ships it. The collection was keyword-and-handle sampled, so its class balance reflects how the data was gathered, not the true prevalence of hate in Twi or Hausa discourse — stated explicitly so no reader infers a base rate from it.

7. Method / Pipeline

1. **Load** Twi and Hausa AfriHate via Hugging Face; use provided splits; hold out the test set untouched.

2. **EDA** — run label.unique() per subset to confirm exact label strings (card documents Hate/Abusive/Normal; the Twi viewer renders "Abuse" — treat loaded values as ground truth), then confirm class counts and per-split test-set sizes (the Normal cell is small in Twi; bin design depends on the real number).
3. **Clean** — minimal: lowercase, strip URLs/handles, normalize whitespace.
4. **Models** — (a) TF-IDF (word + character n-grams) + logistic regression; (b) fine-tuned AfroXLMR-base (fallback AfriBERTa if compute-limited).
5. **Calibration measurement** — for each model × language: macro-F1, then per-class ECE and reliability diagrams using equal-mass (adaptive) bins to cope with small classes; report confidence intervals on ECE.
6. **The blinding quantity** — fraction of each model's errors falling in its high-confidence bins, per class.
7. **Mitigation** — temperature scaling fit on dev; report ECE before/after, overall and per class. Optionally label smoothing (Ahuja et al. found it effective multilingually).
8. **Cross-language comparison** — does the miscalibrated class follow the rare class (Normal in Twi, Hate in Hausa)?
9. **Error analysis** — characterize the confident-but-wrong posts in the worst-calibrated class.

8. Evaluation Plan

Selected: Macro F1 · Per-class precision/recall/F1 · Expected Calibration Error (per class) · Reliability diagrams · Confusion matrix · Error analysis · Bias audit

Accuracy is excluded as a headline: with a 3,278-vs-180 imbalance a majority predictor scores well while detecting nothing. The two headline outputs are (1) the per-class ECE table across both models and both languages, and (2) the reliability diagrams. Success is not a low ECE — it is a clean, defensible answer to whether confidence is equally trustworthy across classes and whether the untrustworthy class follows rarity across languages, with intervals reported so any gap can be judged against the small-sample noise the minority classes introduce. A null result (calibration uniform across classes) is reported as a finding. Temperature scaling is evaluated on whether it fixes minority-class calibration specifically, not just the aggregate — a model whose overall ECE improves while its rare-class ECE does not is exactly the trap the project exists to expose.

9. Literature Grounding

Paper	Problem studied	Method	Dataset	How it supports our project
Guo, Pleiss, Sun & Weinberger (2017). On Calibration of Modern Neural Networks. ICML, PMLR 70:1321–1330. arXiv:1706.04599	Whether modern deep nets are calibrated	ECE, reliability diagrams, temperature scaling	Vision/NLP benchmarks	Supplies our calibration metrics and mitigation, and the finding that deep nets are more overconfident than simpler models — grounding the classical-vs-transformer contrast
Ahuja, Sitaram, Dandapat & Choudhury (2022). On the Calibration of Massively Multilingual Language Models. EMNLP, 4310–4323. DOI 10.18653/v1/2022.emnlp-main.290	Calibration of multilingual models across languages	ECE across languages, temperature scaling, label smoothing	XNLI etc.	Crux: shows miscalibration is worse for low-resource, typologically distant languages — makes the Twi/Hausa question novel and supplies mitigations
Muhammad et al. (2025). AfriHate. NAACL, 1854–1871. DOI	Absent moderation / over-censorship of	Native-speaker annotation, 15	Twitter-derived	Supplies our data, labels, class distributions, and

10.18653/v1/2025.naacl-long.92	African-language hate speech	languages; PLM baselines		published Twi/Hausa baselines
Islam, Seenivasan, Ren & Glocker (2021). Class-Distribution-Aware Calibration for Long-Tailed Recognition. arXiv:2109.05263	Calibration under class imbalance	Class-frequency-aware calibration; shows uniform temperature scaling is suboptimal long-tailed	Long-tailed benchmarks	Grounds the per-class / imbalance core and the choice of adaptive binning and class-aware scaling
Alabi, Adelani, Mosbach & Klakow (2022). Adapting PLMs to African Languages via MAFT (AfroXLMR). COLING, 4336–4349	Adapting XLM-R to African languages	Multilingual adaptive fine-tuning	African corpora	Supplies the transformer baseline for the accuracy-vs-calibration contrast

10. Expected Output

Selected: Notebook(s) · Source code · Figures/tables · Analysis outputs · Data card · Model card · Technical report · Simple demo

A reproducible fixed-seed notebook implementing both models on both languages; a per-class ECE table (2 models × 2 languages × 3 classes); reliability diagrams; a before/after temperature-scaling comparison, overall and per class; the high-confidence-error fraction per class; a data card documenting provenance, licence status, and non-redistribution; a model card stating the classifiers are diagnostic, not deployable; a short technical report; and a triage demo that ranks posts by predicted toxicity and surfaces low-confidence predictions for human review — a direct expression of the finding.

11. Risks and Fallback Plan

Risk	What could fail?	Fallback
Data access / licence	The gated licence turns out to restrict use	We redistribute nothing regardless. If use is restricted, switch to AfriSenti (same languages, CC-licensed sentiment data) and reframe as calibration of sentiment rather than hate classification — same method, same finding shape.
Small classes	Twi Normal (180) and Hate test cells too small for stable per-bin ECE	Equal-mass binning, fewer bins, ECE confidence intervals; lean on Hausa (larger, better-powered) as the primary analysis and treat Twi as the imbalance-stress case.
Model performance	Transformer not more miscalibrated than logistic regression after scaling	That is a finding: “low-resource miscalibration is cheaply recoverable via post-hoc scaling.” Report it.
Time / scope	Two models × two languages × mitigation overruns solo	The per-class ECE table is the project and is non-negotiable. Drop label smoothing first, then the Twi error analysis, then reduce to one model family. The cross-language comparison and temperature scaling stay.